ITA06020 MACHINE LEARNING

Experiment 12 – Iris Flower Classification Using KNN

Aim:

To implement the K-Nearest Neighbours (KNN) algorithm in Python to classify Iris flowers into different species using a real-world dataset.

Dataset: Iris Flower Dataset (available in Scikit-learn).

Algorithm: K-Nearest Neighbours (KNN).

Procedure

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Separate input features and target values.
4. Split the dataset into training and testing data.
5. Normalize the input features.
6. Create a KNN classifier with K = 5.
7. Train the model using training data.
8. Predict the flower species and calculate accuracy.

Python Program

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

data = load_iris()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42, stratify=y

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)
```

```
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Iris Flower Classification using KNN")
print("Training Samples:", len(X_train))
print("Testing Samples:", len(X_test))
print("Predicted Flower:", data.target_names[y_pred[0]])
print("Actual Flower:", data.target_names[y_test[0]])
print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")
```

Output:

```
Iris Flower Classification using KNN
Training Samples: 120
Testing Samples: 30
Predicted Flower: setosa
Actual Flower: setosa
Accuracy: 93.33 %
```

Result

Thus, the K-Nearest Neighbours algorithm was successfully implemented using Python to classify Iris flowers into three different species with an accuracy of 93.33%.